CUBRID: A FOSS RDBMS to Realise Your Next Big Idea

CUBRID is a free and open source relational database management engine. The power of CUBRID lies in its capability to support enterprise level features. It provides certain novel features in the form of object-oriented database element relations, a native middleware broker, and data caching with high performance. Moreover, it has production quality compatibility with many popular databases such as MySQL.

Databases are an integral part of any software application. The persistence that the databases bring in to the applications enable efficient retrievability of relevant data from non-volatile storage. There are a large number of databases such as MySQL, MariaDB, CouchDB, etc. This article provides an overview of a powerful enterprise-ready database management engine, termed CUBRID (pronounced 'cube-rid').

The first appearance of CUBRID was in November 2008. The latest stable release is CUBRID 10.1, which was released in July 2017. It is written in the C language and supports two major operating systems—GNU/Linux and Windows.

The major features of CUBRID are:

- Free and open source
- Powerful
- Offers enterprise features

CUBRID is completely free of cost. As it is open source, you can choose to customise it to your specific requirements. Its enterprise features are listed below.

- **High availability:** The high availability of CUBRID is attributed to its powerful Heartbeat native engine. It works very accurately and has automatic fail-over technology, built in.
- **Globalisation:** CUBRID has excellent I18N (internationalisation) and L10N (localisation) features.
- **Scalability and Big Data optimisation:** CUBRID offers automatic volume expansion with support for multiple volumes.
- **High performance optimised for Web services:** CUBRID has support for multi-threaded and multi-server architecture. It has native broker middleware. The intensive caching feature that CUBRID offers is excellent.

Getting started with CUBRID

This section details the steps required to get started with the CUBRID database management engine. The overall process is illustrated in Figure 3.

Step 1: The first step is to download and install the CUBRID engine. The official Web page provides download links for both GNU/Linux and Windows *(https://www.cubrid.org/downloads/os-select/64-bit/engine)*. You can download the latest stable release of CUBRID 10.1 and execute the installer.

Step 2: After successful installation, you can start the CUBRID service by executing the following command in a terminal:

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cubrid service start
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